Accessible Emotion-Aware E-Learning Platform for **Deaf Students** Studying O/L Information Technology Using **Facial Expression Recognition and an NLP-Based Chatbot**



Problem Statement & Impact

Problem Statement & Impact	Current Issues	Impact / Gravity
Traditional e-learning platforms are mainly designed for hearing students and heavily depend on: • audio explanations • spoken instructions • text-heavy learning materials This creates major learning barriers for deaf O/L ICT students, especially when studying technical concepts	• Students lose attention during online lessons • Existing LMS platforms cannot detect confusion or disengagement • Learning systems provide static learning experiences • Lack of adaptive support for deaf learners • No real-time personalized learning assistance	• Reduced learning engagement • Poor concept understanding • Increased cognitive overload • Learning frustration and fatigue • Educational inequality for deaf students

Person / Group / Society Benefited by the Solution

Secondary Beneficiaries

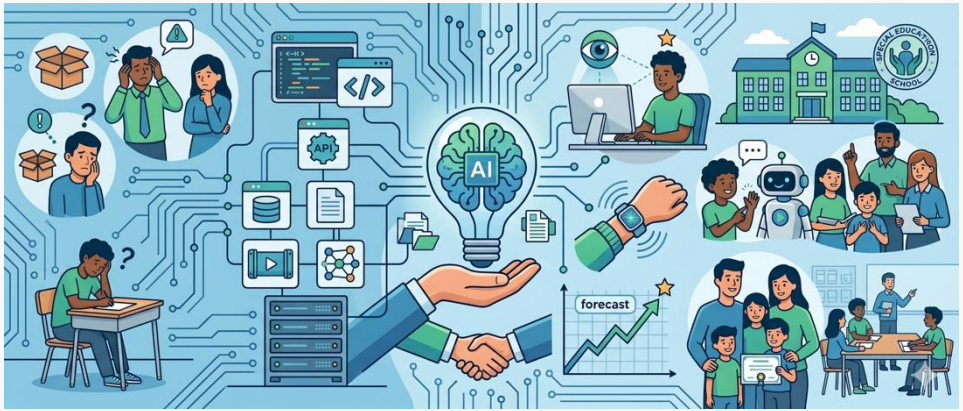
- ICT teachers
- Special education schools
- Inclusive education institutions
- Parents and guardians
- Sri Lankan EdTech sector

Educational & Technological Benefits

- More accessible digital learning
- Personalized adaptive education
- Improved student engagement monitoring
- Better inclusive education support

Primary Beneficiaries

- Deaf and hard-of-hearing O/L ICT students



Design Excellence of the Overall Solution

Key Innovative Features

- Real-time attention monitoring using Computer Vision
- Knowledge Graph-based popup questions
- Emotion-aware adaptive chatbot
- Text-to-sign avatar learning support
- Smart haptic wristband feedback
- Predictive learning analytics
- Full-stack integrated AI learning ecosystem



Technical Excellence

- React + Vite frontend architecture
- FastAPI backend services
- MongoDB-centered database design
- MediaPipe + OpenCV computer vision integration
- WebSocket real-time communication
- Modular AI-assisted learning framework





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Real-Time Emotion & Attention Monitoring Learning Support System

Real-Time Emotion & Attention Monitoring Learning Support System

Problem:

Traditional e-learning systems:

- cannot detect whether students are attentive or distracted.
- fail to identify missed learning segments
- use random popup quizzes
- do not provide adaptive learning reinforcement

Solution:

Real-Time Attention Monitoring

+

Emotion-Aware Learning Analytics

+

Knowledge Graph-Based Popup Learning Support

- Monitors student attention via webcam
- Tracks attention during lessons
- Detects missed learning sections
- Maps lesson content to ICT concepts
- Generates smart popup questions using syllabus links

Features:

- Webcam-based attention monitoring
- Face Mesh attention detection
- EAR & head pose tracking
- Attention log storage
- Missed segment detection
- Knowledge Graph questions
- Diagram learning popups
- GQSA-based question selection
- Answer storage

Real-Time Emotion & Attention Monitoring Learning Support System

User Requirements

- Detect attention and distraction
- Monitor lesson engagement
- Store attention history
- Identify missed segments
- Generate concept-based questions
- Avoid repeated popup questions
- Supports visual learning for deaf students
- Reduces unnoticed learning gaps
- Diagram-assisted learning
- Reduces cognitive overload

Design Excellence / Contribution

- TEPA - Temporal Emotion & Attention Pattern Analysis
  - Analyzes continuous attention behavior
  - Detects distraction periods
  - Stores attention patter
- GQSA - Graph Based Question Selection Algorithm
  - Intelligently selects popup questions
  - Uses current, prerequisite, and related concepts
  - Avoids repeated questions
- Context-Aware Question Selection
- Missed Segment Detection

Real-Time Emotion & Attention Monitoring Learning Support System

User Feedback on Prototype

Initial User Observations:

- popup questions improved lesson engagement
- diagram-supported questions improved understanding
- attention monitoring helped identify concentration loss
- intelligent popup timing improved learning focus

Feedback Highlights:

- Better understanding of difficult ICT concepts
- More interactive than traditional LMS quiz systems
- Helpful for identifying missed lesson content
- Improved adaptive learning support

Technical Stack

Layer	Technology
Frontend	React + Vite
Backend	FastAPI
Database	MongoDB
CV/ML	MediaPipe + OpenCV
Real-Time	WebSockets
Adaptive Logic	TEPA + GQSA



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Adaptive Text-to-Sign Avatar and Transcript-Based Learning Suppor System

Adaptive Text-to-Sign Avatar and Transcript-Based Learning Support System

Problem:

Traditional e-learning systems:

- Written language often acts as a second language, making complex ICT technical jargon difficult to process.
- Most platforms lack real-time sign language translations or sign-based visualizations for dynamic content.
- There is a severe lack of replay mechanisms and visual aids tailored specifically for the linguistic needs of deaf learners.

Solution:

- Dual-Coding Integration:

To reduce cognitive load, the system utilizes the Visual Concept Panel.

- Spatial-Linguistic Pairing:

The panel listens to the avatar's current gesture and generates a high-quality 3D isometric diagram of the signed concept, pairing sign language with concrete pictorial representation.

Features:

- Parses complex English into simplified Sign Language glosses
- Smoothly interpolates between 21-point MediaPipe hand poses and upper-body gestures using mathematical vector transitions
- Automatically renders 3D educational diagrams (e.g., LAPTOP, PROJECTOR) synced with the avatar
- Automatically switches to fingerspelling when specific sign gestures are unavailable in the database.

Adaptive Text-to-Sign Avatar and Transcript-Based Learning Support System

User Requirements

- **Accessibility First:**
Accurate conveyance of ICT terminology without lag.

- **Unobscured Viewing:**
UI elements (text boxes/concept images) must never overlap with the avatar's hand or facial movements.

- **Cognitive Support:**
Reinforce learning through simultaneous visual aids (Dual-Coding).

- **Robustness:**
Graceful handling of API rate limits (e.g., Gemini 429 errors) with a maximum delay of only a few seconds.

Design Excellence / Contribution

❖ **Glassmorphism**

- Uses translucency (backdrop-blur) and subtle inner borders to create depth while separating panels from the Avatar Stage.

❖ **Dynamic Layout**

- The Visual Concept panel is a floating element in the top-right, ensuring the avatar's arm movements remain completely unobstructed.

❖ **Micro-interactions**

- Features smooth fade-in animations, active-state neon glows, and sophisticated typography for a polished user experience.

Adaptive Text-to-Sign Avatar and Transcript-Based Learning Support System

Visual Concept Architecture

❖ Trigger

- React detects the avatar signing a technical term (e.g., SCANNER).

❖ Request

- 2. An asynchronous request is sent to the FastAPI backend.
- 3. Prompt Engineering: FastAPI utilizes Gemini 2.5 Flash to draft a descriptive prompt for a 3D isometric diagram.

❖ Failsafe

- If Gemini is unavailable, the system falls back to a hardcoded high-quality template prompt.

❖ Generation & Caching

- Pollinations.ai generates the image; the backend downloads and saves the file to /public/concept_images/ for static serving.

❖ Display

- The frontend receives the local URL and fades the image into the panel.

Adaptive Text-to-Sign Avatar and Transcript-Based Learning Support System

User Feedback on Prototype

Initial User Observations:

- Students preferred sign-supported explanations
- Replay support improved understanding
- Fingerspelling fallback improved unknown word handling
- Visual sign playback increased engagement

Feedback Highlights:

- Better accessibility for deaf learners
- Improved visual understanding of ICT lessons
- More interactive than text-only learning systems
- Helpful for self-paced revision

Technical Stack

Layer	Technology
Frontend	React + Vite
Backend	FastAPI
Database	MongoDB
Visualization	Landmark-based gesture rendering
APIs	Axios + REST APIs



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NLP-Based Emotion-Aware, Intent-Aware Adaptive Chatbot

NLP-Based Emotion-Aware, Intent-Aware Adaptive Chatbot

Problem:

Traditional educational chatbots:

- provide static answers
- ignore student emotional state
- cannot adapt explanations
- do not support deaf learners effectively
- lack curriculum-aware assistance

Solution:

- detects learning intent (Learning Mode / Exam Mode)
- adapts explanations based on student learning state
- provides simplified and visual-friendly support
- integrates with ICT syllabus structure
- supports adaptive learning guidance

Features:

- Learning Mode vs Exam Mode detection
- Emotion-aware response adaptation (EARA)
- Knowledge Graph prerequisite awareness
- Dynamic answer style adaptation
- Chatbot session storage
- FastAPI backend integration
- React frontend chatbot interface

NLP-Based Emotion-Aware, Intent-Aware Adaptive Chatbot

User Requirements

- students can ask ICT-related questions
- system provides adaptive responses
- chatbot supports exam preparation
- system gives simplified explanations
- student learning state influences responses
- chatbot supports concept revision
- Clear and concise explanations
- reduced cognitive overload
- visual learning support integration
- beginner-friendly answer structure
- O/L ICT syllabus alignment
- prerequisite topic awareness
- learning reinforcement support

Design Excellence / Contribution

- EARA - Emotion Aware Response Adaptation

  The chatbot changes explanation style based on:
  - Understanding, boredom, distraction, confusion

- Answer Compression Mode (Exam Mode Detection)
- Dynamic Difficulty Escalation
- Concept Re-Entry Detection
- Optional Micro-Challenge with Instant Summary Access
- Forgetting Curve-Based Reinforcement (Login Quiz Mode)
- Repeated Query Alert Algorithm
- Lesson Understanding Analytics Dashboard

NLP-Based Emotion-Aware, Intent-Aware Adaptive Chatbot

User Feedback on Prototype

Initial User Observations:

- Students preferred simplified explanations
- Exam mode answers improved revision speed
- Adaptive responses increased engagement
- Short focused answers helped distracted learners

Feedback Highlights:

- Easier understanding of difficult ICT concepts
- Better support for self-learning
- More interactive than normal chatbot systems
- Useful for exam preparation

Technical Stack

Layer	Technology
Frontend	React + Vite
Backend	FastAPI
Database	MongoDB
Adaptive Logic	EARA
APIs	Axios + REST APIs



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Smart Haptic Wristband and Predictive Learning Analytics System

Smart Haptic Wristband and Predictive Learning Analytics System

Problem:

Current e-learning systems for deaf students:

- Provide only visual alerts
- Cannot privately notify distracted learners
- Lack personalized haptic feedback systems
- Lack smart notification alerts

Solution:

- Sends haptic vibration alerts through a smart wristband
- Use haptic vibrations and LED notification
- Customizable haptic vibrations
- Visual notification on wristband

Features:

- ESP32-based wristband integration
- Real-time vibration alert system
- OLED notification support
- React frontend + FastAPI backend integration

Smart Haptic Wristband and Predictive Learning Analytics System

User Requirements

- Send vibration alerts to students
- Detect distraction or disengagement
- Private tactile feedback for deaf learners
- Non-audio learning assistance
- Reduced interruption during lessons
- Improve learning focus
- Better engagement monitoring

❖ ***Vibration Examples***

- Short Pulse
- Double Pulse
- Long Pulse
- Emergency Pulse

Design Excellence / Contribution

❖ Tactile Learning Feedback

- Converts learning states into vibration patterns
- Provides private haptic learning support
- Custom vibration patterns
- Control notification alerts

❖ ***Supported Alert Types***

- Distraction Alert
- Chatbot Reply
- Popup Question
- Exam Reminder

❖ ***OLED Message Examples***

- FOCUS BACK
- CHAT REPLY
- NEW QUESTION
- EXAM TIP

Smart Haptic Wristband and Predictive Learning Analytics System

Future Improvements

- Compact wearable casing
- Battery optimization
- Cloud deployment
- Real-user testing

Expected Outcomes

- Improved learning focus
- Helpful private feedback mechanism
- Better engagement
- Innovative assistive learning support

Current Hardware Prototype

- ESP32 mini board
- OLED display
- vibration motor
- push button
- breadboard setup

Technical Stack

Layer	Technology
Frontend	React + Vite
Backend	FastAPI
Database	MongoDB
IoT Device	ESP32 Wristband
APIs	Axios + REST APIs

Adaptive Emotion-Aware E-Learning Platform for Deaf O/L ICT Students

Customer Persona

Primary Users

- Deaf and hard-of-hearing O/L ICT students
- Special education students
- Students in inclusive classrooms

Secondary Users

ICT teachers
Educational institutions
Parents and guardians
Inclusive education schools



Market Size

Global Perspective

- Over 430 million people worldwide experience disabling hearing loss (WHO)

- Growing demand for:
  - accessible e-learning
  - adaptive AI education systems
  - assistive educational technologies

Sri Lankan Perspective

- Increasing adoption of digital learning platforms
- Limited accessibility-focused AI learning systems
- High demand for inclusive educational support

Adaptive Emotion-Aware E-Learning Platform for Deaf O/L ICT Students

What is Unique in Our Solution?

Unlike traditional LMS systems, our platform combines:

- Real-time attention monitoring
- Knowledge Graph-based adaptive learning
- Emotion-aware intelligent chatbot
- Text-to-sign avatar support
- Smart haptic wristband feedback
- Adaptive learning analytics

Proposed Revenue Model

- School Subscriptions - Monthly or annual licensing
- SaaS Educational Platform - Cloud-based student access
- Government / NGO Partnerships - Inclusive education initiatives
- Premium Features - Advanced analytics, Personalized reports

Sustainability of the Solution

Technical Uniqueness

Integrated AI + Accessibility Approach

Combines:

- Computer Vision
- Adaptive Learning
- Knowledge Graphs
- NLP-based assistance
- Sign-supported visualization
- IoT wearable feedback

Designed for:

• Deaf O/L ICT students

Long-Term Expansion : Other school subjects, Sinhala/Tamil support, Mobile learning apps, Full AI tutoring systems

THANK YOU !